



Discrete Structures

- Chapter 2, Part 2, Functions
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 - Spring 2026
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Ordered tuples

The ordered n -tuple $(a_1, a_2, \dots, a_n)$ is an ordered collection of n objects

- Unlike sets order matters here
- Notation: $(a_1, a_2, \dots, a_n)$ unlike $\{a_1, a_2, \dots, a_n\}$
- Repetition matters too
- More like C++/Java arrays, except for type restriction is not applied

Ordered tuples

The ordered n -tuple $(a_1, a_2, \dots, a_n)$ is an ordered collection of n objects

$(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ means $a_i = b_i$ for $1 \leq i \leq n$

- $(1, 2, 9) = (1, 2, 9)$
- $(1, \text{apple}, 8, \text{car}) \neq (1, \text{apple}, \text{car}, 8)$
- $(3, 5, 7) \neq (3, 5, 7, 11)$

Which one of the following is true?

ICP 5-1 $(3, 5, 7, 11) = (3, 7, 5, 11)$

a) True b) False

ICP 5-2 $(3, 5, 5, 7) \neq (3, 5, 7)$

a) True b) False

ICP 5-3 $(3, 5, 5, 7) = (3, 5, 7, 5)$

a) True b) False

ICP 5-4 $(x, A) \neq (A, x)$

a) True b) False

Cartesian Product

Ordered 2-tuples ($n = 2$) are called ordered pairs

Cartesian product of sets A and B is the set of all ordered pairs (x, y) , where $x \in A$ and $y \in B$

$$A \times B = \{(x, y) \mid (x \in A) \wedge (y \in B)\}$$

▷ Also known as the cross product

$$S = \{x_1, x_2, x_3\} \text{ and } G = \{A, B\}$$

$$S \times G = \{(x_1, A), (x_1, B), (x_2, A), (x_2, B), (x_3, A), (x_3, B)\}$$

$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$: the Cartesian plane or Euclidean Plane

▷ Cartesian product of $\mathbb{R}$ (x-axis) and $\mathbb{R}$ (y-axis)

Cartesian Product

$$X = \{1, 2\},$$

$$Y = \{a, b, c\}$$

- $X \times Y = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$
- $Y \times X = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$
- $X \times X = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$
- $X \times \emptyset = \emptyset$
- $\emptyset \times X = \emptyset$

ICP 5-5

Generally, $A \times B = B \times A$?

a) True

b) False

$A \times B \neq B \times A$ unless $A = \emptyset$ or $B = \emptyset$ or $A = B$

Cartesian Product of more than two sets

Readily generalize to more than 2 sets

$$A = \{p, q\}, \quad B = \{i, j\}, \quad C = \{4, 5, 6\}$$

$$A \times B \times C = \left\{ (p, i, 4), (p, i, 5), (p, i, 6), (p, j, 4), (p, j, 5), (p, j, 6), \right. \\ \left. (q, i, 4), (q, i, 5), (q, i, 6), (q, j, 4), (q, j, 5), (q, j, 6) \right\}$$

Cardinality of Cartesian product is the product of cardinalities

$$|A_1 \times A_2 \times \dots \times A_n| = |A_1| \times |A_2| \times \dots \times |A_n|$$

Functions

Let X and Y be two sets.

A function f maps **each** element of X to **exactly one** element of Y

$$\text{Let } S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}, \quad G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$$

- Crazy Professor's map: $\{(x_1, F), (x_2, F), (x_3, F), (x_4, F), (x_5, F)\}$
- Another crazy map: $\{(x_1, A), (x_2, A), (x_3, A), (x_4, A), (x_5, A)\}$
- Reasonable map: $\{(x_1, C), (x_2, A), (x_3, B), (x_4, B), (x_5, A)\}$

Functions

Let X and Y be two sets.

A function f maps **each** element of X to **exactly one** element of Y

- $f : X \mapsto Y$
- $f(x) = y$, if x is mapped to y ; y is assigned to x
- Each $x \in X$ is mapped to exactly one $y \in Y$
- A $y \in Y$ may be assigned to many elements in X

$f : X \mapsto Y$ is a subset of $X \times Y$, such that for every $x \in X$, f contains exactly one ordered pair with first component x

Function Graphical Representation

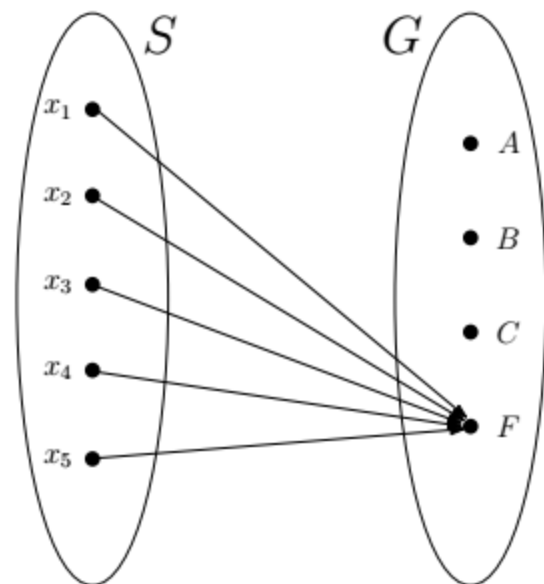
- $f : X \mapsto Y$ can be represented graphically (bipartite graphs)
- 'Draw' sets X and Y . For $f(x) = y$ draw arrow from $x \in X$ to $y \in Y$

$$S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}$$

$$G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$$

Crazy Professor's map:

$$\{(x_1, F), (x_2, F), (x_3, F), (x_4, F), (x_5, F)\}$$



Function Graphical Representation

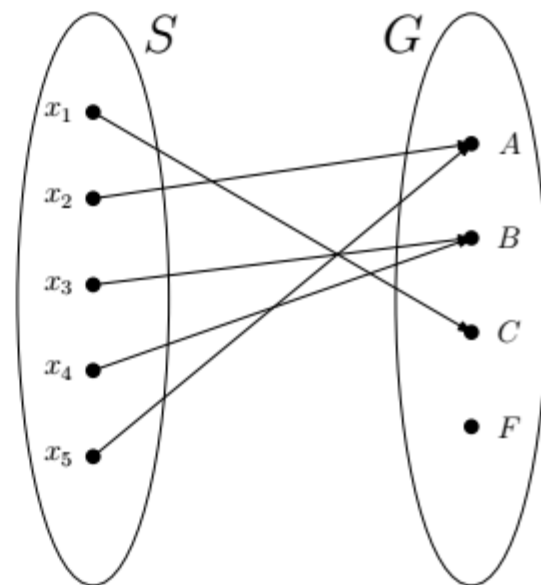
- $f : X \mapsto Y$ can be represented graphically (bipartite graphs)
- 'Draw' sets X and Y . For $f(x) = y$ draw arrow from $x \in X$ to $y \in Y$

$$S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}$$

$$G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$$

Reasonable map:

$$\{(x_1, C), (x_2, A), (x_3, B), (x_4, B), (x_5, A)\}$$



Functions: Basic Terms

Let X and Y be two sets.

A function f maps **each** element of X to **exactly one** element of Y

Let $f : X \mapsto Y$ and let $f(x) = y$

- X is the domain of f
- Y is the codomain of f
- y is the image of x
- x is the pre-image of y
- **Range of f :**
set of images of all elements of X

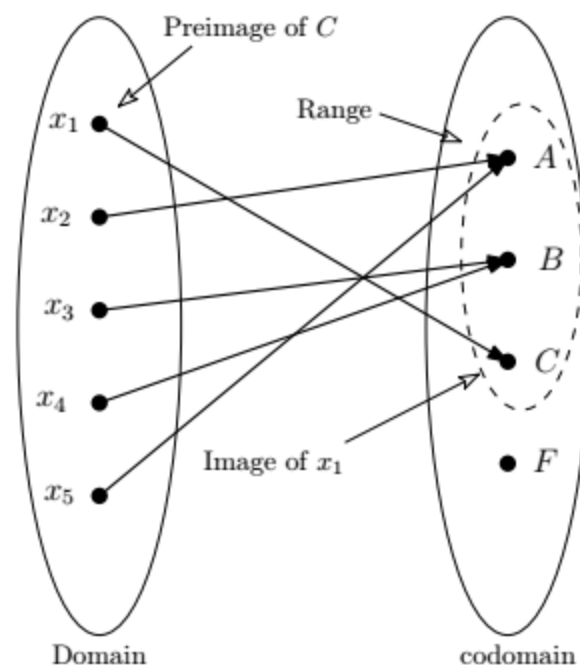
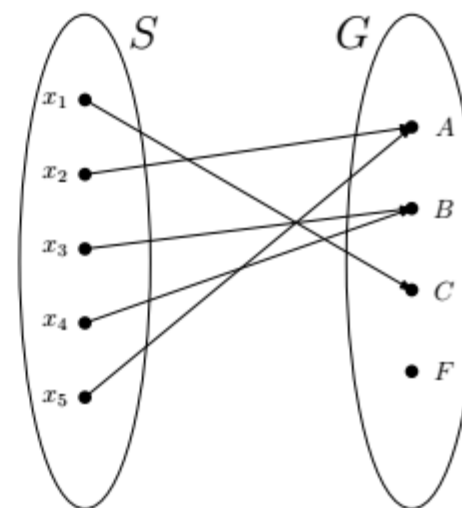


Figure: $f : X \mapsto Y$

Function: Basic Terms

- Let $S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}$, $G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$
- $f : S \mapsto G := \{(x_1, C), (x_2, A), (x_3, B), (x_4, B), (x_5, A)\}$

- $f(x_1) = C$
- $f(x_2) = A$
- $f(x_3) = B$
- $f(x_4) = B$
- $f(x_5) = A$



ICP 5-5 Range of $f = \{A, B, C\}$?

a) True

b) False

Function: Basic Terms

- Let $S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}$, $G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$
- $f : S \mapsto G := \{(x_1, F), (x_2, F), (x_3, F), (x_4, F), (x_5, F)\}$

Function: Basic Terms

■ Let $S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}$, $G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$

■ $f : S \mapsto G := \{(x_1, F), (x_2, F), (x_3, F), (x_4, F), (x_5, F)\}$

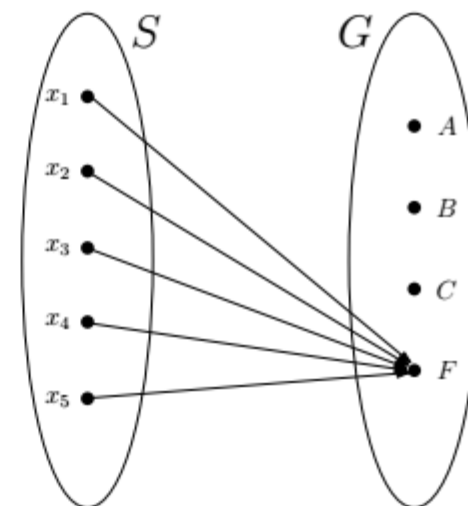
■ $f(x_1) = F$

■ $f(x_2) = F$

■ $f(x_3) = F$

■ $f(x_4) = F$

■ $f(x_5) = F$



ICP 5-6 Range of $f = \{F\}$?

a) True

b) False

Function: Basic Terms

- Let $S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}$, $G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$
- $f : S \mapsto G := \{(x_1, C), (x_2, A), (x_4, B), (x_5, A)\}$

Function: Basic Terms

■ Let $S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}$, $G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$

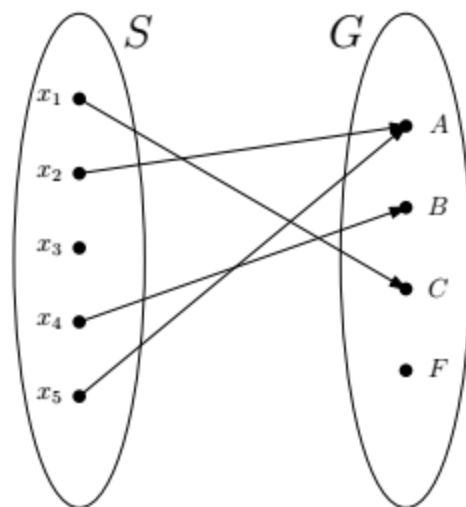
■ $f : S \mapsto G := \{(x_1, C), (x_2, A), (x_4, B), (x_5, A)\}$

■ $f(x_1) = C$

■ $f(x_2) = A$

■ $f(x_3) = B$

■ $f(x_4) = A$



ICP 5-7

Range of $f = \{A, B, C\}$?

a) True

b) False

Function: Basic Terms

- Let $S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}$, $G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$
- $f : S \mapsto G := \{(x_1, C), (x_2, A), (x_3, B), (x_4, B), (x_5, A), (x_4, C)\}$

Function: Basic Terms

■ Let $S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}, \quad G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$

■ $f : S \mapsto G := \{(x_1, C), (x_2, A), (x_3, B), (x_4, B), (x_5, A), (x_4, C)\}$

■ $f(x_1) = C$

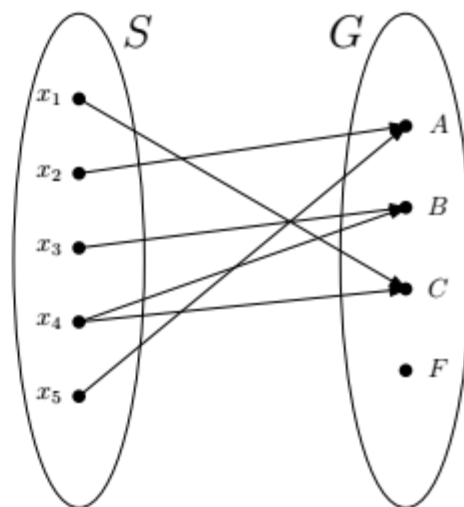
■ $f(x_2) = A$

■ $f(x_3) = B$

■ $f(x_4) = B$

■ $f(x_5) = A$

■ $f(x_4) = C$



ICP 5-8 Range of $f = \{A, B, C\}$?

a) True

b) False

Functions: Mapping Rule

For large domains difficult to describe functions with graphs or sets

Functions can be specified by a mapping rule (formula)

$$\text{Let } S = \underbrace{\{x_1, x_2, x_3, x_4, x_5\}}_{\text{Students}}, \quad G = \underbrace{\{A, B, C, F\}}_{\text{Grades}}$$

$f : S \mapsto G$ is defined as follows:

$$f(x_i) = \begin{cases} A & \text{if } \text{score}(x_i) \geq 90\% \\ B & \text{if } 80\% \leq \text{score}(x_i) < 90\% \\ C & \text{if } 60\% \leq \text{score}(x_i) < 80\% \\ F & \text{if } \text{score}(x_i) < 60\% \end{cases}$$

Functions: Mapping Rule

Let $f : \mathbb{Z} \mapsto \mathbb{Z}$ be given as

$$f(x) = x^2$$

- Domain of f
- Codomain of f
- Image of -3 and 5
- Preimage of $25, 4, -3, 30$
- Range of f

Functions: C++/JAVA

```
function SIGN( $x$ )           ▷ returns int
  if  $x < 0$  then
    return -1
  else if  $x = 0$  then
    return 0
  else
    return 1
```

- Domain of funct()
- Codomain of funct()
- Image of $-3, 2, 5, 0$
- Preimage of $0, 4, 1, -1$
- Range of funct()

Types of functions: One-to-One

A function $f : X \mapsto Y$ is **one-to-one** (or **injective**) iff

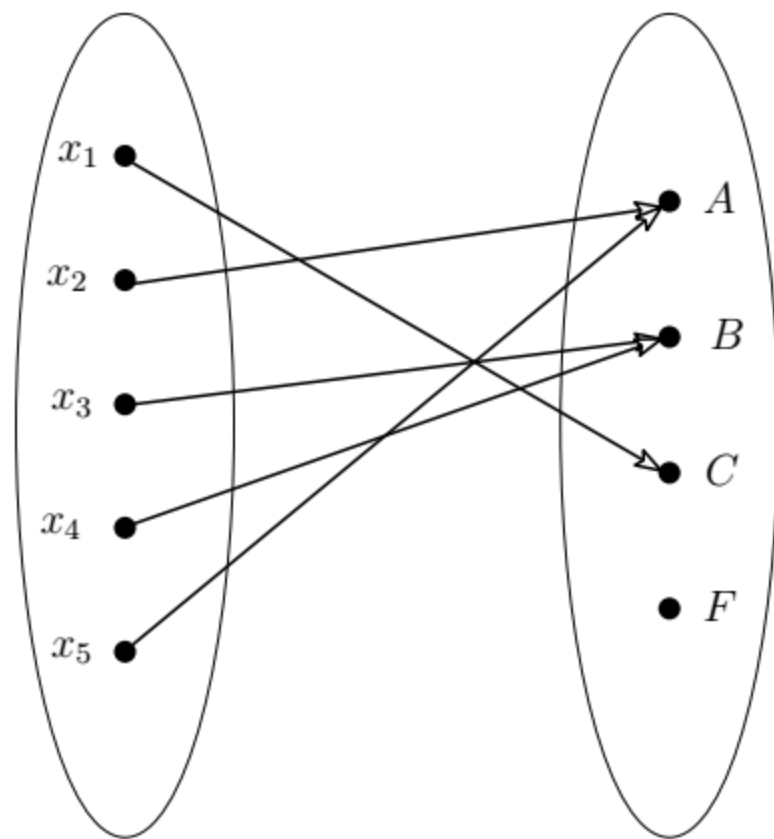
$$\forall x_1, x_2 \in X \quad (f(x_1) = f(x_2) \rightarrow x_1 = x_2)$$

Each element of X is mapped to a unique element of Y

Think of the contrapositive

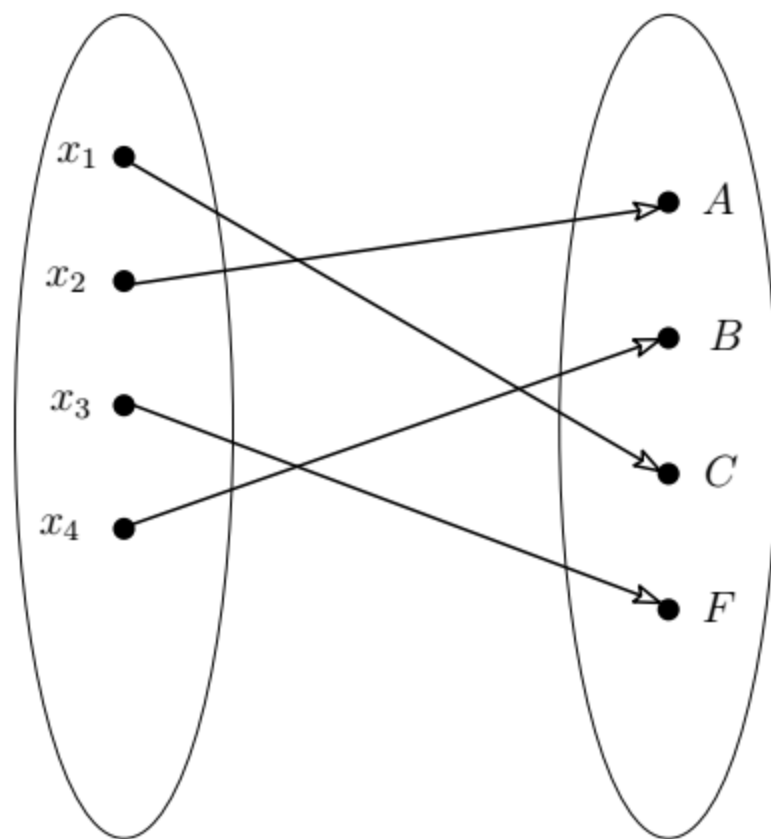
$$\forall x_1, x_2 \in X \quad (x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2))$$

Types of functions: One-to-One



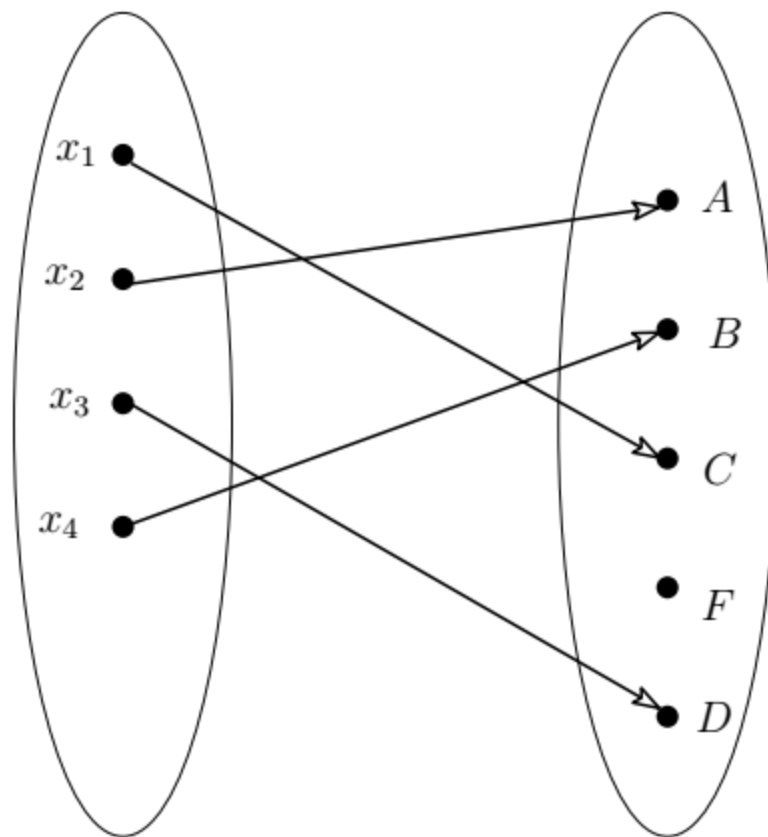
Not one-to-one
 $f(x_2) = f(x_5)$

Types of functions: One-to-One



one-to-one

Types of functions: One-to-One



one-to-one

Types of functions: One-to-One

Let $f : \mathbb{Z} \mapsto \mathbb{Z}$ be

$$f(x) = 2x - 3$$

Is f one-to-one?

$$f(x_1) = f(x_2) \rightarrow 2(x_1) - 3 = 2(x_2) - 3 \rightarrow x_1 = x_2$$

Hence, f is one-to-one

Types of functions: One-to-One

Which of the following functions is one-to-one?

ICP 5-9

$$f : \mathbb{Z} \mapsto \mathbb{R}, f(x) = x^2$$

a) True

b) False

ICP 5-10

$$f : \mathbb{R} \mapsto \mathbb{R}, f(x) = x^3$$

a) True

b) False

ICP 5-11

$$f : \mathbb{Z} \mapsto \mathbb{Z}, f(x) = 2x$$

a) True

b) False

ICP 5-12

$$f : \mathbb{Z} \mapsto \mathbb{N}, f(x) = |x|$$

a) True

b) False

ICP 5-13

$$f : \text{people} \mapsto \text{people}, \\ f(x) = \text{father of } x$$

a) True

b) False

Types of functions: Onto

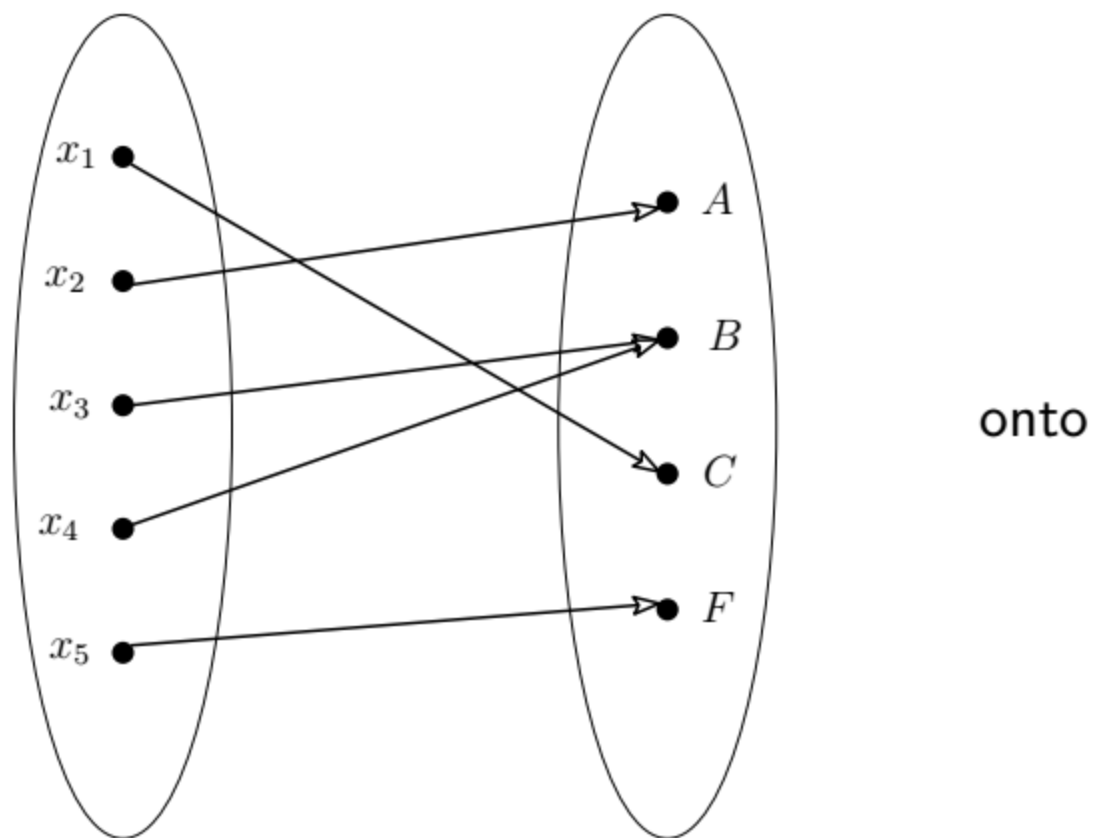
A function $f : X \mapsto Y$ is **onto** (or **surjective**) iff
for every element $y \in Y$ there is an element $x \in X$ with $f(x) = y$

Each element of Y is assigned to some element of X

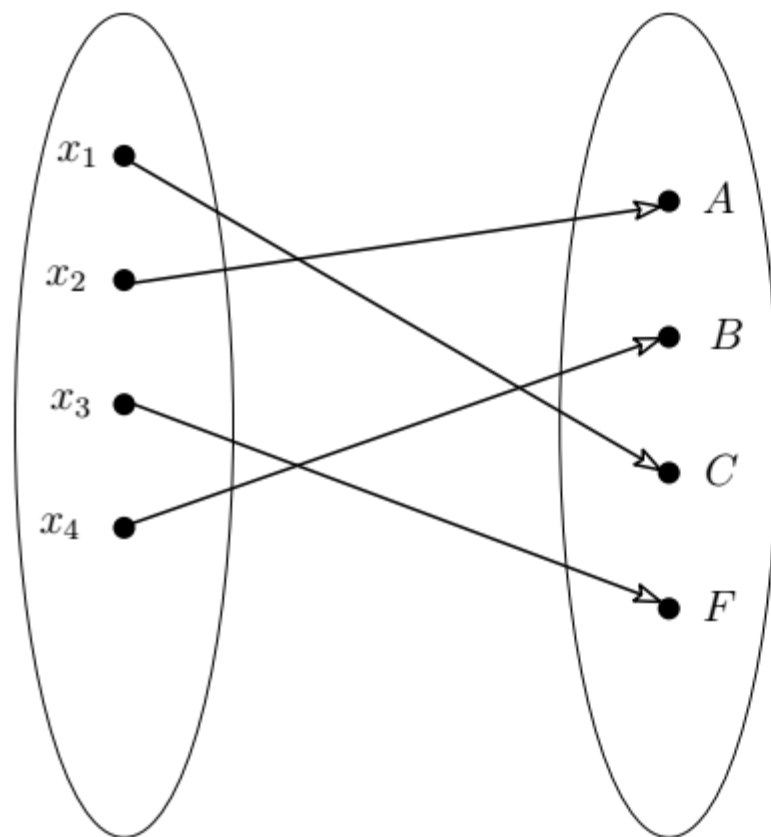
$$\forall y \in Y \exists x \in X \ f(x) = y$$

Range = codomain

Types of functions: Onto

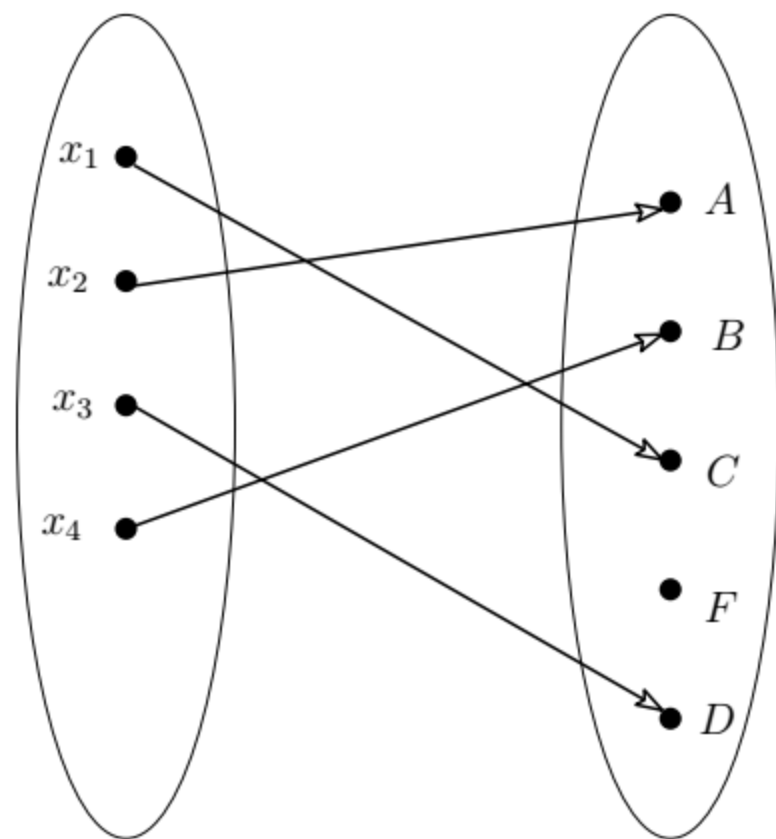


Types of functions: Onto



onto

Types of functions: Onto



not onto

Types of functions: Onto

Which of the following functions is onto?

ICP 5-14

$$f : \mathbb{Z} \mapsto \mathbb{R}, f(x) = x^2$$

a) True

b) False

ICP 5-15

$$f : \mathbb{R} \mapsto \mathbb{R}, f(x) = x^3$$

a) True

b) False

ICP 5-16

$$f : \mathbb{Z} \mapsto \mathbb{Z}, f(x) = 2x$$

a) True

b) False

ICP 5-17

$$f : \mathbb{Z} \mapsto \mathbb{N}, f(x) = |x|$$

a) True

b) False

ICP 5-18

$$f : \text{people} \mapsto \text{people}, \\ f(x) = \text{father of } x$$

a) True

b) False

Types of functions: Onto

Let $f : \mathbb{Z} \mapsto \mathbb{Z}$ be

$$f(x) = 2x - 3$$

What is the range of f ? Is f onto?

We characterize $range(f)$ (a set) from the definition of f and check if it equals codomain

$$\begin{aligned} y \in range(f) &\leftrightarrow y = 2x - 3 && x \in \mathbb{Z} \\ &\leftrightarrow y = 2(x - 2) + 1 \\ &\leftrightarrow y \text{ is odd} \end{aligned}$$

$$range(f) \neq codomain(f)$$

Hence, f is not onto

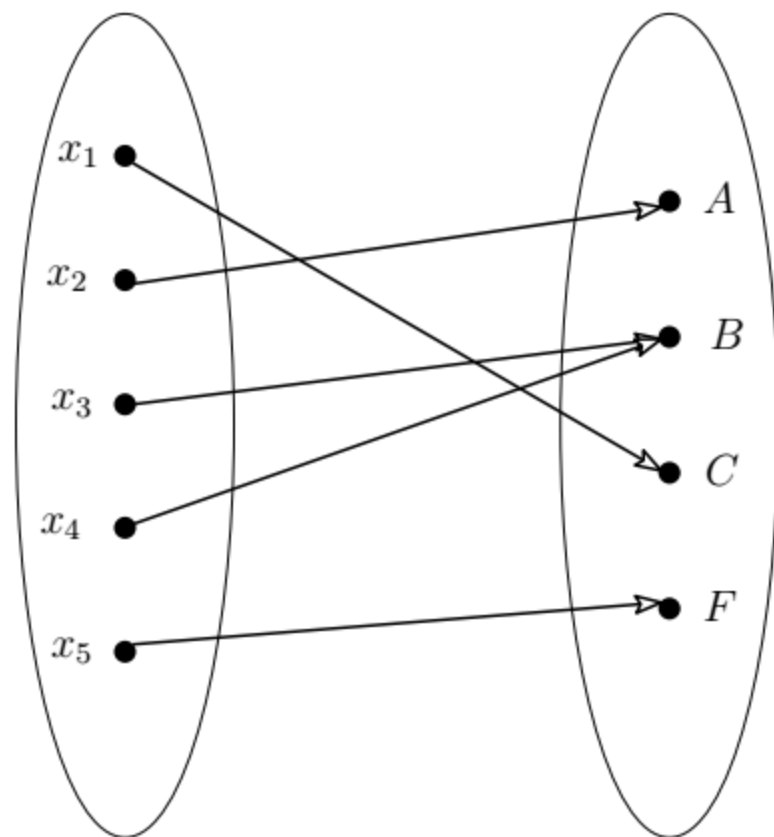
Types of functions: Bijection

A function $f : X \mapsto Y$ is **one-to-one correspondence** (or **bijective**) iff
it is **both one-to-one** and **onto**

- Each element of X is mapped to a unique element of Y
- Each element of Y is assigned to some element of X

- if X and Y are finite sets, then $|X| = |Y|$
- **$|\text{domain}| = |\text{codomain}|$**

Types of functions: Bijection

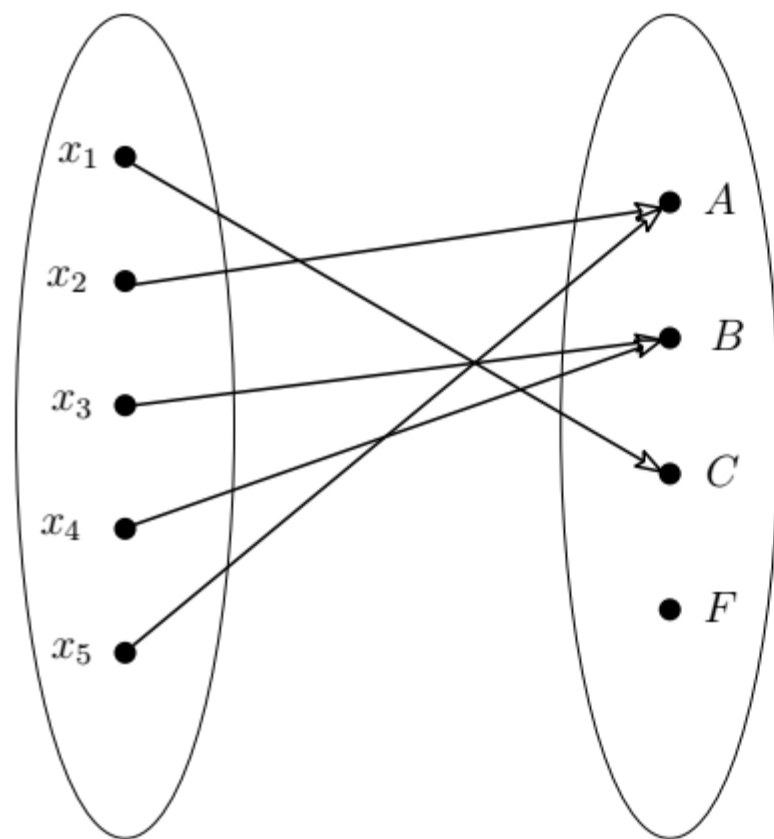


Not bijection

onto

not one-to-one

Types of functions: Bijection

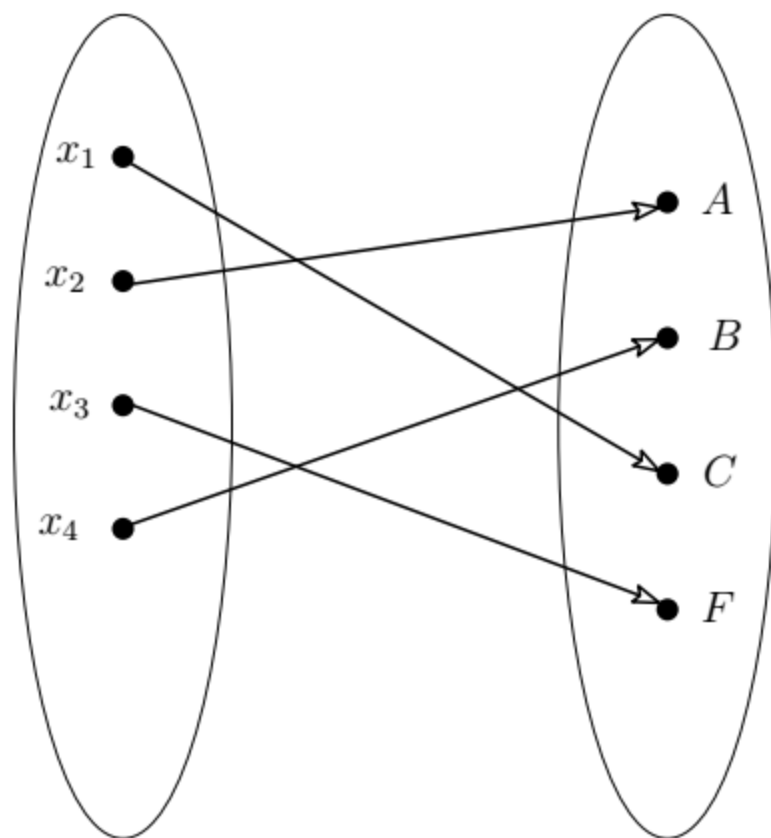


Not bijection

not onto

not one-to-one

Types of functions: Bijection

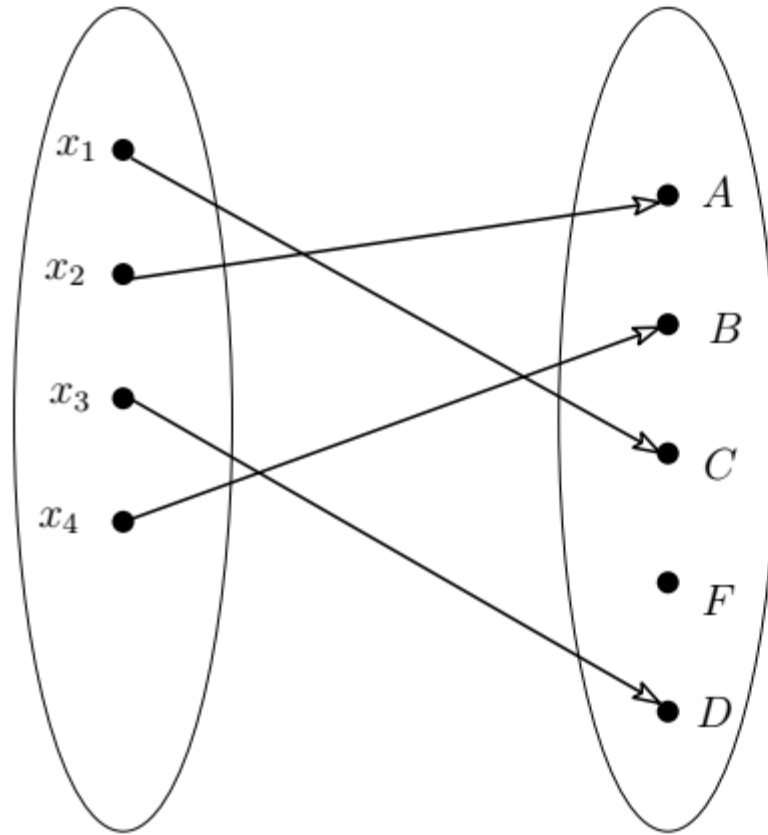


bijection

onto

one-to-one

Types of functions: Bijection



Not bijection

not onto

one-to-one

Types of functions: Bijection

Which of the following functions is a bijection?

ICP 5-19 $f : \mathbb{Z} \mapsto \mathbb{R}, f(x) = x^2$

a) True b) False

ICP 5-20 $f : \mathbb{R} \mapsto \mathbb{R}, f(x) = x^3$

a) True b) False

ICP 5-21 $f : \mathbb{Z} \mapsto \mathbb{Z}, f(x) = 2x$

a) True b) False

ICP 5-22 $f : \mathbb{Z} \mapsto \mathbb{N}, f(x) = |x|$

a) True b) False

ICP 5-23 $f : \text{people} \mapsto \text{people},$
 $f(x) = \text{father of } x$

a) True b) False

Types of functions: Bijection

Let $f : \mathbb{R} \mapsto \mathbb{R}$ be

$$f(x) = 2x - 3$$

Is f a bijection?

- For any $y \in \mathbb{R}$, since $y+3/2 \in \mathbb{R}$ and $f(y+3/2) = y$

▷ so f is onto

- $f(x_1) = f(x_2) \rightarrow 2(x_1) - 3 = 2(x_2) - 3 \rightarrow x_1 = x_2$

▷ so f is one-to-one

- Hence, f is a bijection

Types of functions: Cardinalities

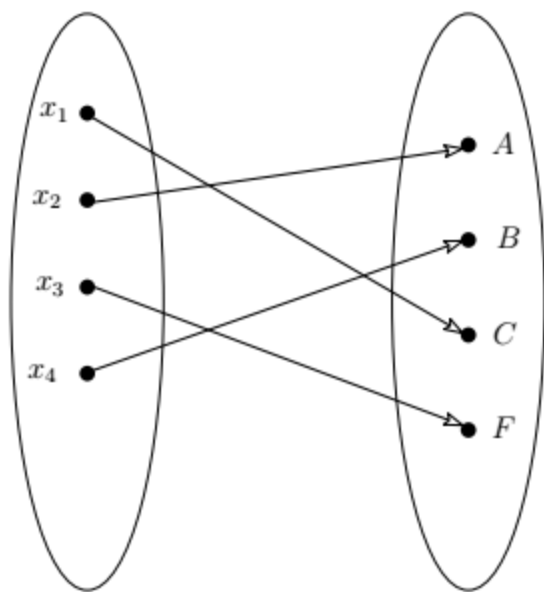
If $f : X \mapsto Y$ is a bijection and X and Y are finite sets, then $|X| = |Y|$

What can we conclude about $|X| \oslash |Y|$,

- 1 when f is one-to-one?
- 2 when f is onto?

Inverse of a function

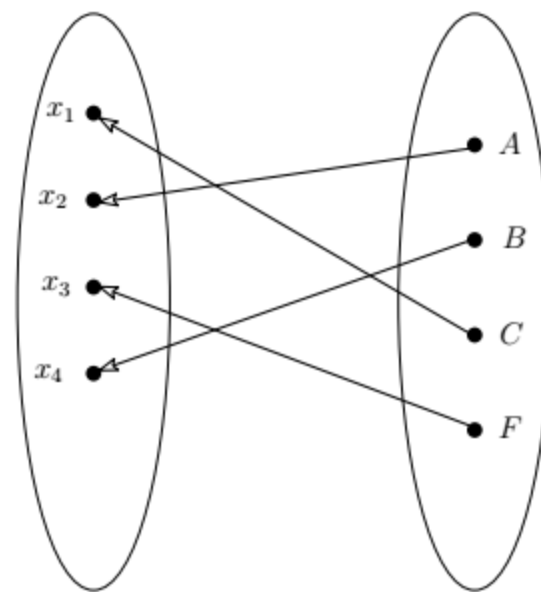
If $f : X \mapsto Y$ is a bijection, then if we reverse the arrows we get a bijection



bijection

onto

one-to-one

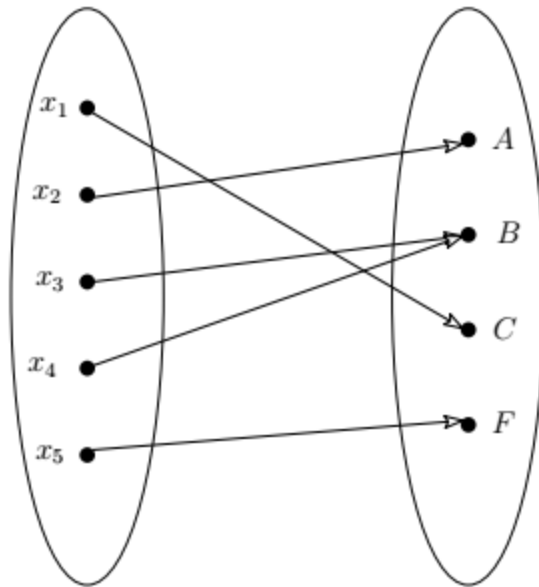


Inverse of a function

If $f : X \mapsto Y$ is a bijection, then if we reverse the arrows we get a bijection

ICP 5-24

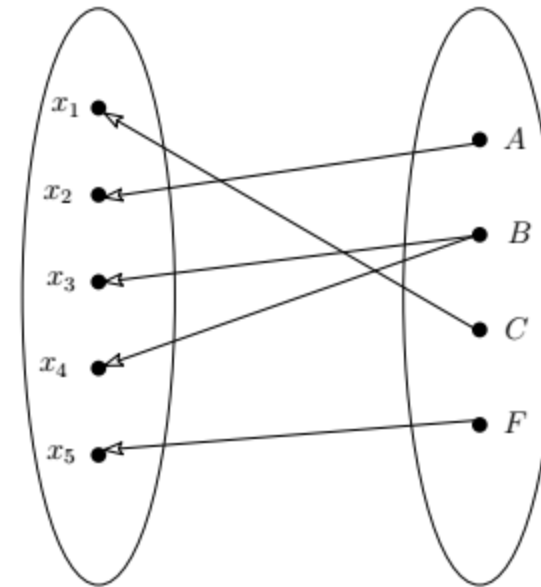
Can we say this when f is not one-to-one? a) Yes b) No



Not bijection

onto

not one-to-one



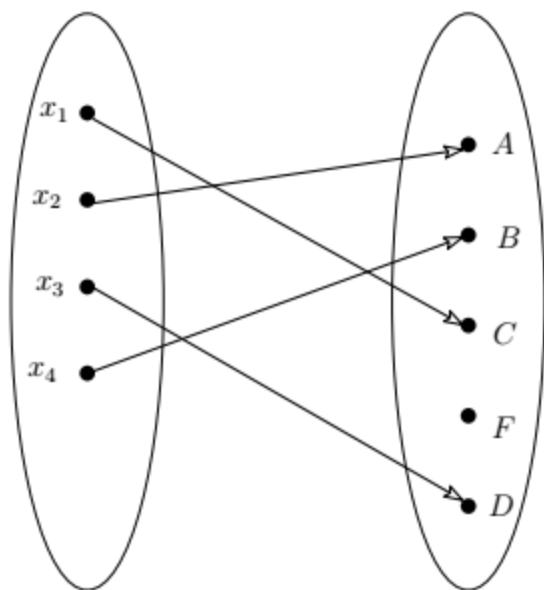
Inverse of a function

If $f : X \mapsto Y$ is a bijection, then if we reverse the arrows we get a bijection

ICP 5-25

Can we say this when f is not onto?

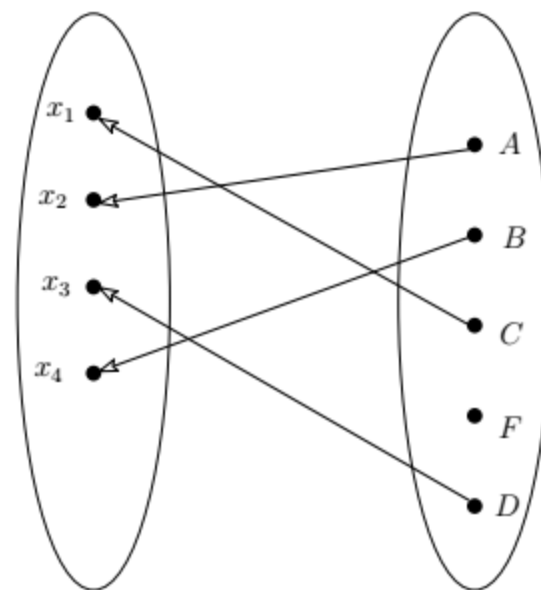
a) Yes b) No



Not bijection

not onto

one-to-one



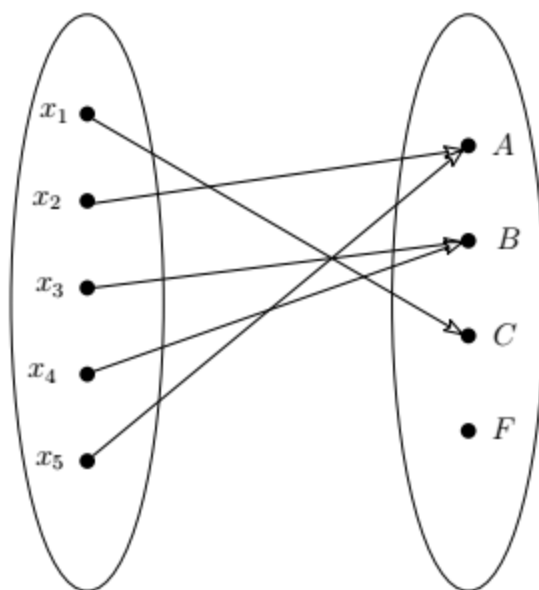
Inverse of a function

If $f : X \mapsto Y$ is a bijection, then if we reverse the arrows we get a bijection

ICP 5-26

Can we say this when f is neither?

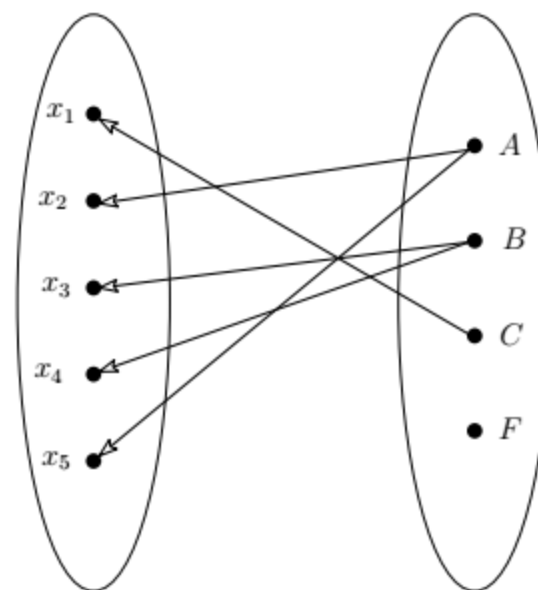
a) Yes b) No



Not bijection

not onto

not one-to-one



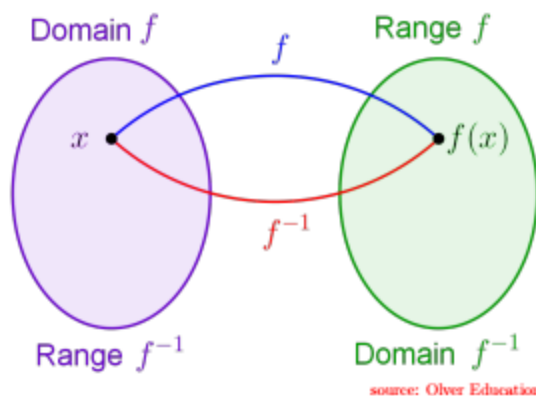
Inverse of function

If $f : X \mapsto Y$ is a bijection, then if we reverse the arrows we get a bijection

If $f : X \mapsto Y$ is a bijection, $f^{-1} : Y \mapsto X$ is the inverse function such that

$$f^{-1}(b) = a, \text{ when } f(a) = b$$

If an inverse exists, then $f(a) = b \Leftrightarrow f^{-1}(b) = a$



Inverse of function

Let $f : \mathbb{R} \mapsto \mathbb{R}$ be

$$f(x) = 2x - 3$$

Is f a bijection? What is f^{-1} ?

For any $y \in \mathbb{R}$, since $y+3/2 \in \mathbb{R}$ and $f(y+3/2) = y$

▷ so f is onto

$$f(x_1) = f(x_2) \rightarrow 2(x_1) - 3 = 2(x_2) - 3 \rightarrow x_1 = x_2$$

▷ so f is one-to-one

Hence, f is a bijection

Suppose $f^{-1}(y) = x$

Let $y = 2x - 3$ and solve for x , we get

$$f^{-1}(y) = y+3/2$$

Inverse of function

ICP 5-27

Let $f : \mathbb{R}^- \mapsto \mathbb{R}^+$ be

$$f(x) = x^2$$

Is f a bijection? What is f^{-1} ?

$$\mathbb{R}^- = \{x \in \mathbb{R} \mid x \leq 0\} \quad \text{and} \quad \mathbb{R}^+ = \{x \in \mathbb{R} \mid x \geq 0\}$$

$$f^{-1}(y) = -\sqrt{y}$$

Composing functions

The output of one function can be used as input to another function

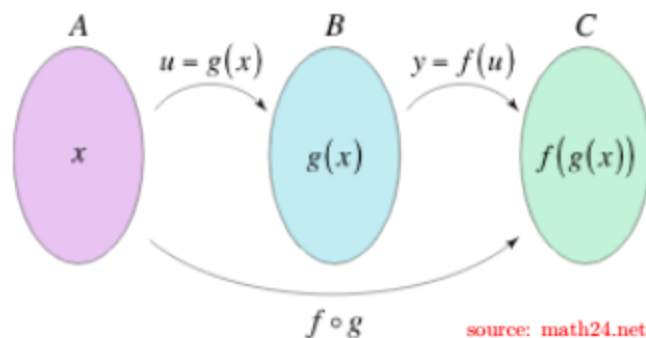
▷ **Restriction on range and domain**

Let $g : A \mapsto B$ and let $f : B \mapsto C$

The composition of the functions f and g is a function, $(f \circ g) : A \mapsto C$ is

$$(f \circ g)(a) = f(g(a))$$

- The output of g is input to f
- Range of g (actual outputs) must be a subset of domain of f



Composition

■ Let $f : \mathbb{R} \mapsto \mathbb{R}$ be $f(x) := x^3$

■ Let $g : \mathbb{Z} \mapsto \mathbb{Z}$ be $g(x) := x^2$

■ $(f \circ g)(x) : \mathbb{Z} \mapsto \mathbb{R}$

$$(f \circ g)(x) = f(g(x)) = f(x^2) = (x^2)^3 = x^6$$

What will be $(g \circ f)$?

Range of inner function must be a subset of domain of outer function

Composition

■ Let $f : \mathbb{Z} \mapsto \mathbb{Z}$ be $f(x) := x^3$

■ Let $g : \mathbb{Z} \mapsto \mathbb{Z}$ be $g(x) := x^2$

■ $(f \circ g)(x) : \mathbb{Z} \mapsto \mathbb{Z}$

$$(f \circ g)(x) = f(g(x)) = f(x^2) = (x^2)^3 = x^6$$

■ $(g \circ f)(x) : \mathbb{Z} \mapsto \mathbb{Z}$

$$(g \circ f)(x) = g(f(x)) = g(x^3) = (x^3)^2 = x^6$$

Composition

- Let $f : \mathbb{Z} \mapsto \mathbb{Z}$ be $f(x) := x^2$
- Let $g : \mathbb{Z} \mapsto \mathbb{Z}$ be $g(x) := 2x$
- **ICP 5-28** $(f \circ g)(x) : \mathbb{Z} \mapsto \mathbb{Z}$

$$(f \circ g)(x) = f(g(x)) = f(2x) = (2x)^2 = 4x^2$$

- **ICP 5-29** $(g \circ f)(x) : \mathbb{Z} \mapsto \mathbb{Z}$

$$(g \circ f)(x) = g(f(x)) = g(x^2) = 2(x^2) = 2x^2$$

$(f \circ g)(\cdot)$ is not necessarily the same as $(g \circ f)(\cdot)$

Sometimes, the other side of composition may not even be possible

Composing function with itself

A function can be composed with itself

Let $f : \mathbb{Z} \mapsto \mathbb{Z}$ be

$$f(x) = x + 1$$

$$(f \circ f)(x) = f(f(x)) = f(x + 1) = (x + 1) + 1 = x + 2$$

Composing function with itself repeatedly

A function can be composed with itself multiple times

$f : \mathbb{Z} \mapsto \mathbb{Z}$ be

$$f(x) = x + 1$$

$$(f \circ (f \circ f))(x) = f(f(f(x))) = f(f(x + 1)) = f(x + 2) = x + 3$$

Let $father : People \mapsto People$ be defined as

$$father(x) = \text{father of } x$$

$$father(father(father(x))) = \text{great grand father of } x$$

Composing function with it's inverse

When a function is invertible (when it's a bijection), then it can be composed with it's inverse

$$\text{If } f = g^{-1}, \text{ then } (f \circ g)(x) = f(g(x)) = g^{-1}(g(x)) = x$$

Let $f : \mathbb{Z} \mapsto \mathbb{Z}$ be $f(x) = x + 1$

Let $g : \mathbb{Z} \mapsto \mathbb{Z}$ be $g(x) = x - 1$

$$\triangleright f = g^{-1}$$

$$f \circ g(x) = f(g(x)) = f(x - 1) = (x - 1) + 1 = x$$

Monotonic Functions

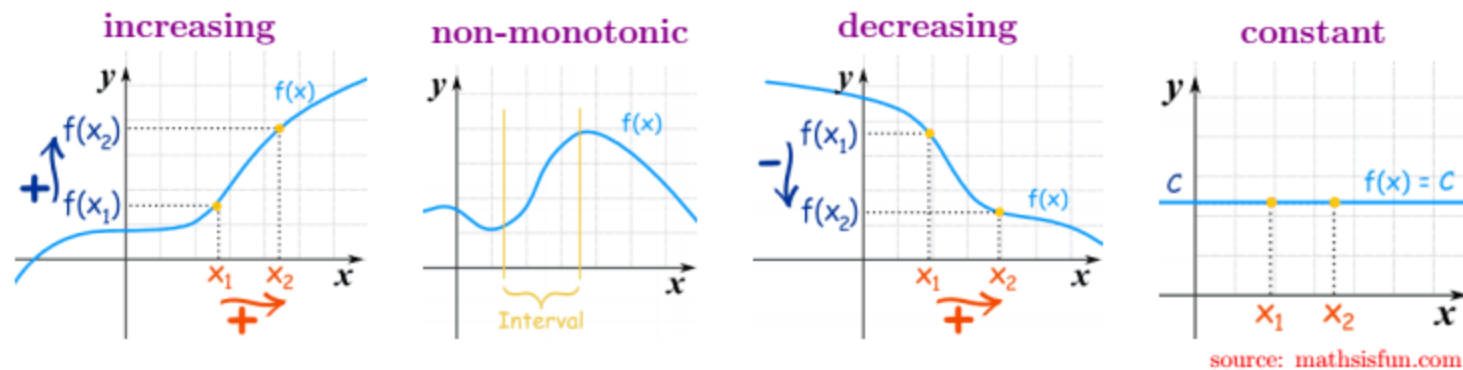
A function $f : \mathbb{R} \mapsto \mathbb{R}$ is **increasing** if $f(x) \leq f(y)$ whenever $x < y$

$$\forall x \forall y \ (x < y \rightarrow f(x) \leq f(y))$$

f is called **strictly increasing** if $f(x) < f(y)$ when $x < y$

▷ **Decreasing** and **strictly decreasing** functions are defined similarly

Collectively, such functions are called **monotonic functions**



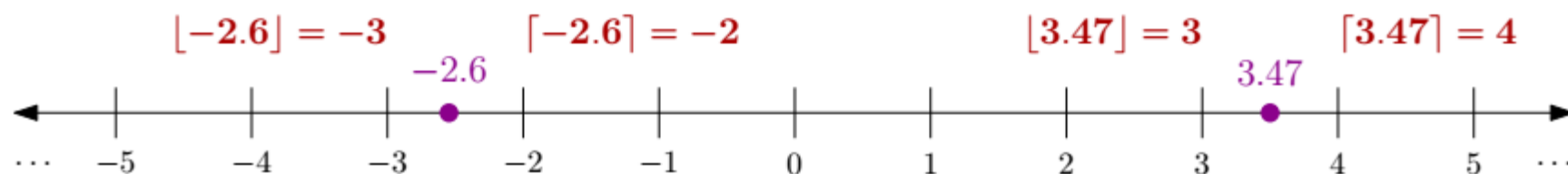
Note: strictly increasing/decreasing functions are necessarily one-to-one

Important Numeric Functions: Ceiling and Floors

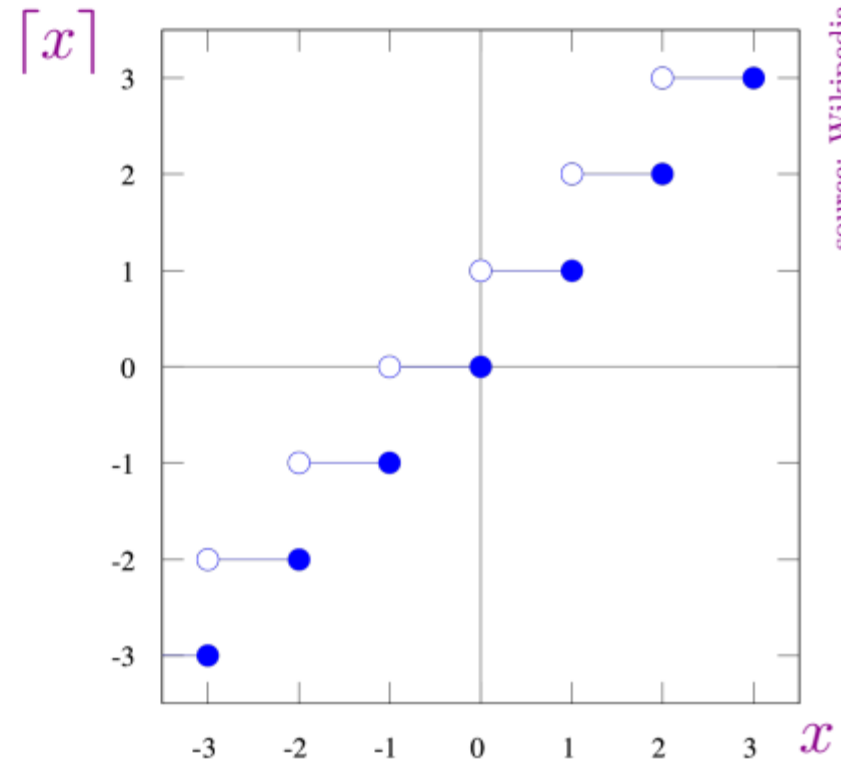
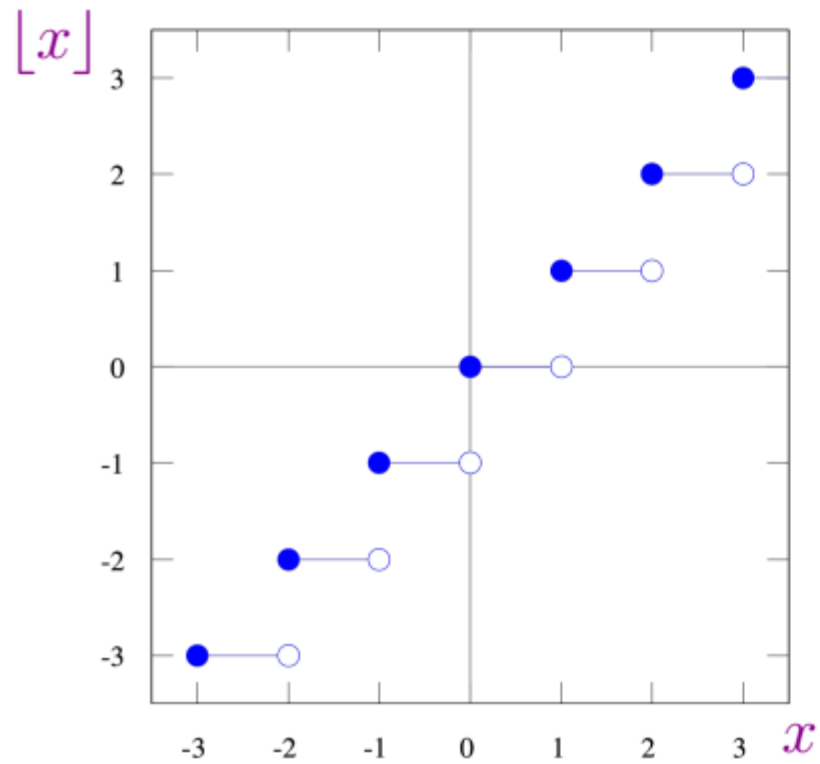
The floor and ceiling functions map real numbers to integers ($\mathbb{R} \mapsto \mathbb{Z}$)

Given $x \in \mathbb{R}$,

- **floor**(x), $\lfloor x \rfloor$ is the largest integer $\leq x$
- **ceiling**(x), $\lceil x \rceil$ is the smallest integer $\geq x$



Important Numeric Functions: Ceiling and Floors



source: Wikipedia

Important Numeric Functions: Ceiling and Floors

ICP 5-30

- $\lfloor 1.7 \rfloor, \lceil 1.7 \rceil$
- $\lfloor 3.2 \rfloor, \lceil 3.2 \rceil$
- $\lfloor 3.5 \rfloor, \lceil 3.5 \rceil$
- $\lfloor 3.9 \rfloor, \lceil 3.9 \rceil$
- $\lfloor 11 \rfloor, \lceil 11 \rceil$

ICP 5-31

- $\lfloor -1.7 \rfloor, \lceil -1.7 \rceil$
- $\lfloor -3.2 \rfloor, \lceil -3.2 \rceil$
- $\lfloor -3.5 \rfloor, \lceil -3.5 \rceil$
- $\lfloor -3.9 \rfloor, \lceil -3.9 \rceil$
- $\lfloor -11 \rfloor, \lceil -11 \rceil$